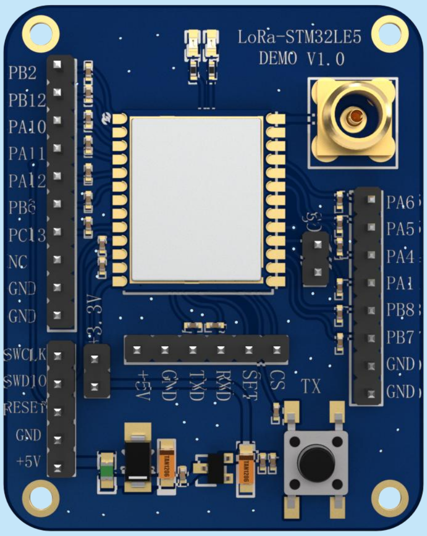
LoRa-STM32WLE5 SOC Wireless Module Evaluation Board

Product Specification

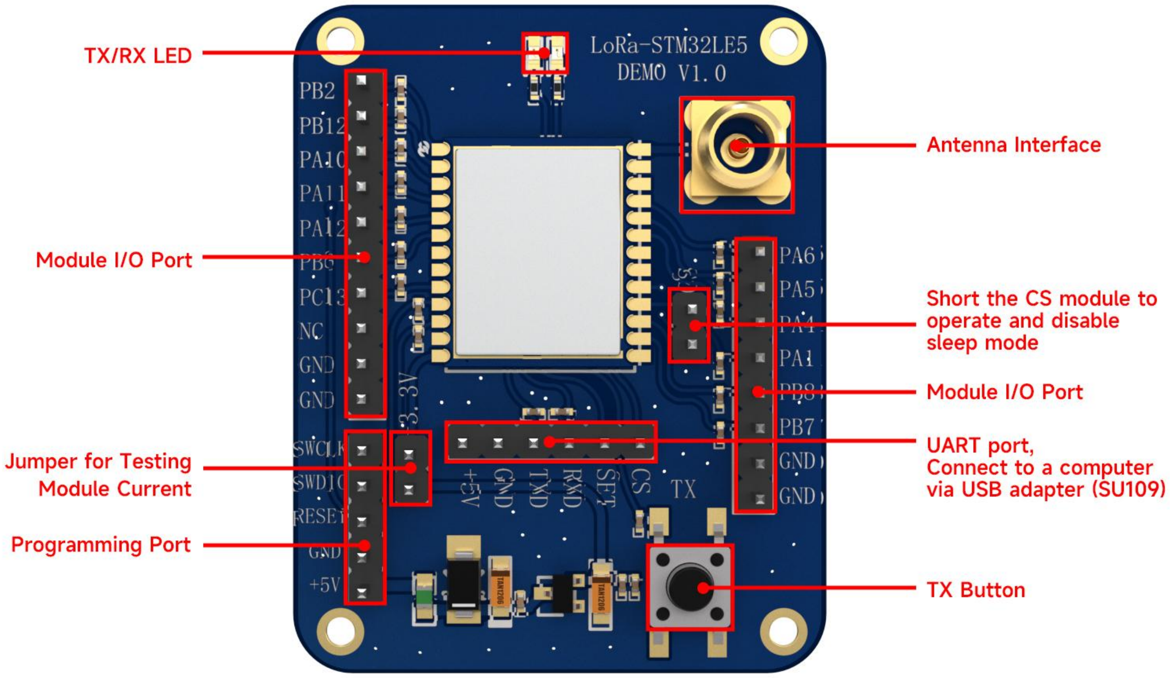


Catalogue

Note: Revision History

Revision	Date	Comment
V1.0	2024-5	First release

1. Internal Block Diagram



Internal block diagram: CS pin pulled low for operation, disconnected for sleep.

2. Example Program Description

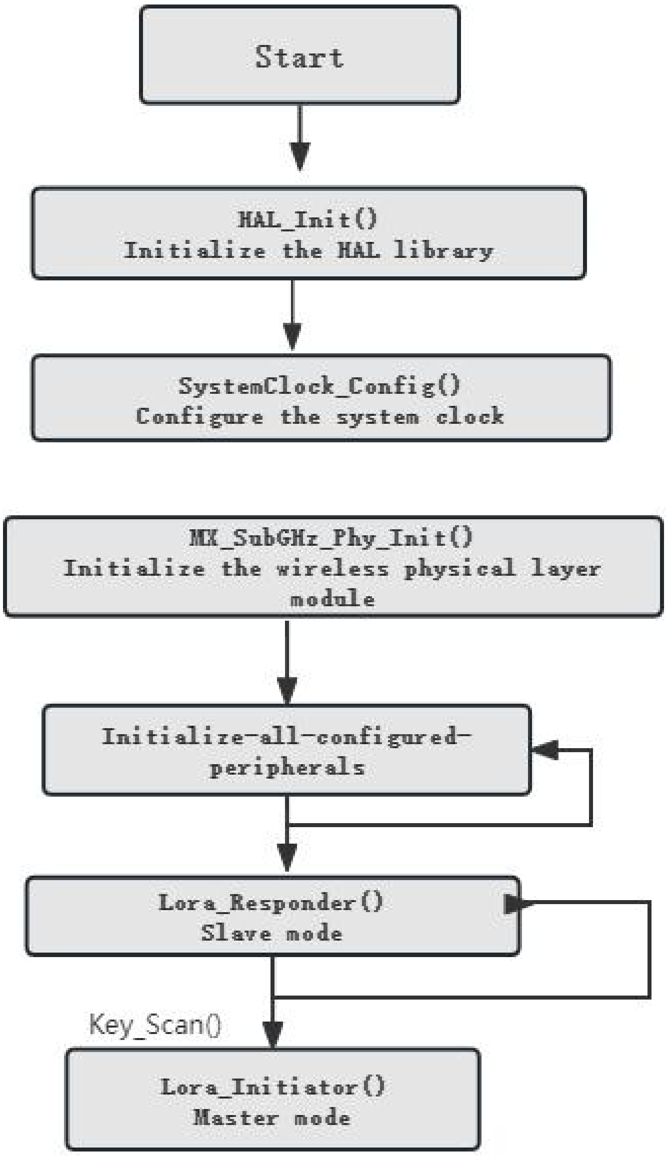
2.1 Introduction

The communication example program is a simple LoRa transceiver routine between two LoRa-STM32WLE5 boards. After powering on, the modules are in a waiting state to receive (slave device). The module with the TX button pressed acts as the master device and starts sending data. It sends a "send_test" message and then waits for a reply. The device that receives the "send_test" message becomes the slave device and replies with a "return_test" message. Once the master device receives the "return_test" message, it continues to send the "send_test" message. This establishes a continuous transmission and reception process.

Under normal circumstances, when the two boards establish a communication relationship, the red LED (transmit LED) and the blue LED (receive LED) will flash. When the two boards are fully synchronized, i.e., when the RX windows of both boards are synchronized (both the red and blue LEDs are off) or when the TX windows of both boards are synchronized (the red LED is flashing), the communication relationship cannot be established. In this case, simply restart one of the boards.

2.2 Flowchart

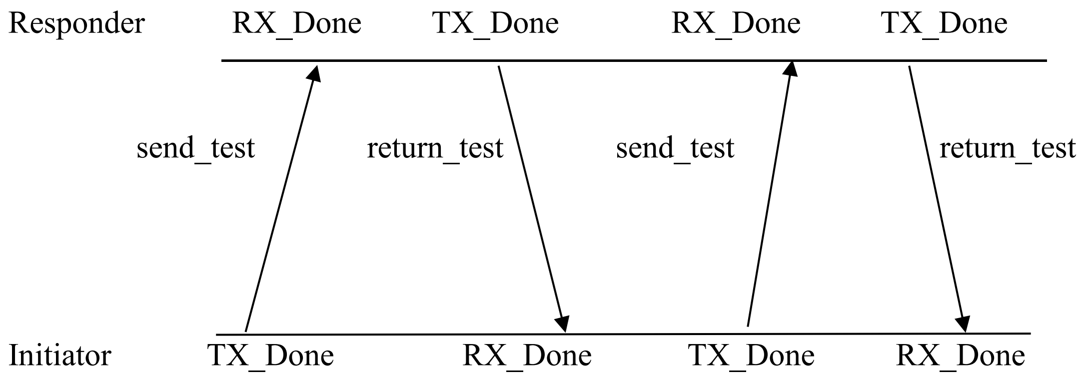
1) Main Process



Refer to main.c.

The MX_SubGHz_Phy_Init() function calls SystemApp_Init() to complete the configuration of LoRa parameters. It first initializes LoRa and registers the callback function Radio.Init (&RadioEvents). Then, it configures the transmission parameters with Radio.SetTxConfig(), the reception parameters with Radio.SetRxConfig(), sets the payload length with Radio. SetMaxPayloadLength(), and configures the operating frequency with Radio.SetChannel().

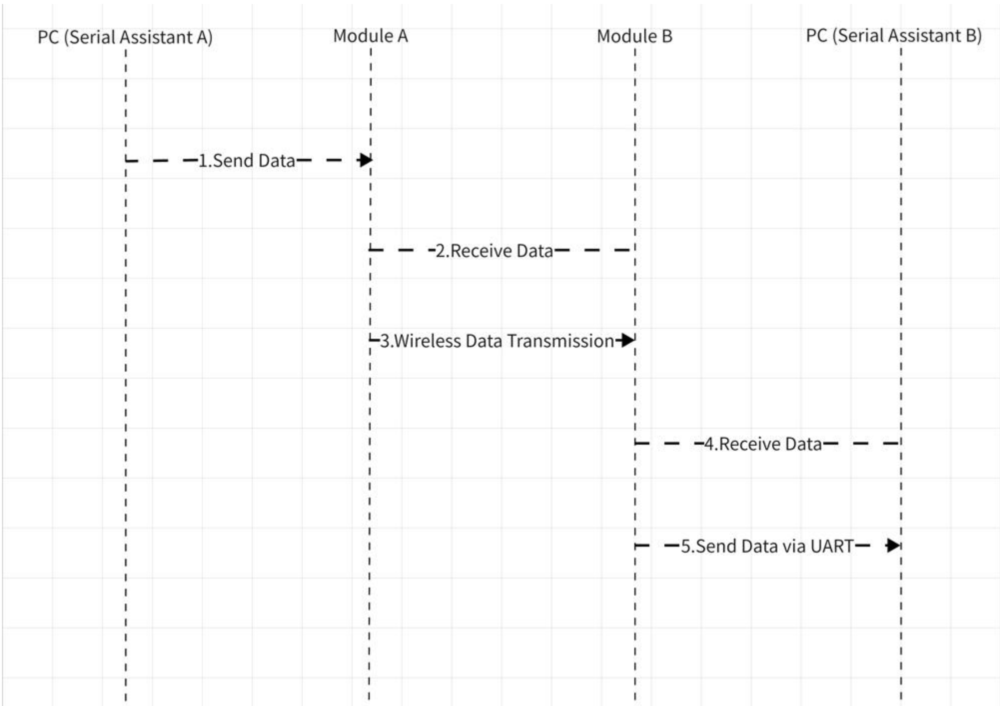
2) **Communication Process**



3. UART Functionality

(1) Data Transmission Process

Use the serial port data transmission mode, requiring a power reset. This function allows users to send data to a wireless module via a serial debugging assistant. After receiving the data, the module sends it to another module. The received information is then displayed on the serial debugging assistant of the receiving module.



(2) UART Parameters Configuration

- Baud Rate: 115200 bps
- Data Bits: 8
- Stop Bits: 1
- Parity: None

(3) UART Data Packet Format Each data packet includes the actual data followed by a carriage return (newline) character as the termination symbol. The detailed format is as follows: <Data><Carriage Return>

4. Parameter Configuration

Category	Parameter	**Value**	**Description**
TX/RX Configuration	RF_FREQUENCY	433000000	Transmission Frequency
	RX_TIMEOUT_VALUE	0	Reception Timeout Value
	TX_TIMEOUT_VALUE	1000	Transmission Timeout Value
Lora SetTXConfig Parameters	TX_OUTPUT_POWER	22	Transmission Power
	LORA_BANDWIDTH	500 KHz	Signal Bandwidth
	LORA_SPREADING_FACTOR	7	Spreading Factor
	LORA_CODINGRATE	7	Coding Rate
	LORA_PREAMBLE_LENGTH	8	Preamble Length
	LORA_SYMBOL_TIMEOUT	5	Symbol Timeout
	LORA_FIX_LENGTH_PAYLOAD_ON	false	Fixed Payload Length
	LORA_IQ_INVERSION_ON	false	IQ Inversion

5.Pin Description

Signal	GPIO	**Software Definition**
TX Button	PB4	KEY_TX_Pin
Receive Indicator	PB5	LED_RX_Pin
Transmit Indicator	PB3	LED_TX_Pin
CS Pin	PA0	CS_Pin